

Lab 3(BCS551)-DML Commands

Objective: At the end of assignments ,participants will be able to understand basic DML,Insert rows using insert command ,Update using update command and delete using delete command

Select Commands:

Query1:

```
Select *  
FROM HR.DEPARTMENTS;
```

Output1:

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	Administration	200	1700
2	20	Marketing	201	1800
3	30	Purchasing	114	1700
4	40	Human Resources	203	2400
5	50	Shipping	121	1500
6	60	IT	103	1400
7	70	Public Relations	204	2700
8	80	Sales	145	2500

Query2:

```
Select department_id,location_id  
FROM HR.DEPARTMENTS;
```

Output2:

	DEPARTMENT_ID	LOCATION_ID
1	10	1700
2	20	1800
3	30	1700
4	40	2400
5	50	1500
6	60	1400
7	70	2700
8	80	2500

Query3:

Select last_name,salary,salary+300
FROM HR.EMPLOYEES;

Output3:

Download ▾ Execution time: 0.006 seconds			
	LAST_NAME	SALARY	SALARY+300
1	King	24000	24300
2	Yang	17000	17300
3	Garcia	17000	17300
4	James	9000	9300
5	Miller	6000	6300
6	Williams	4800	5100
7	Jackson	4800	5100
8	Nguyen	4200	4500

Operator Precedence:

Query4:

Select last_name,salary,12*salary+100
FROM HR.EMPLOYEES;

Output4:

Download ▾ Execution time: 0.002 seconds			
	LAST_NAME	SALARY	12*SALARY+100
1	King	24000	288100
2	Yang	17000	204100
3	Garcia	17000	204100
4	James	9000	108100
5	Miller	6000	72100
6	Williams	4800	57700
7	Jackson	4800	57700
8	Nguyen	4200	50500

Query5:

Select last_name,salary,12*(salary+100)
FROM HR.EMPLOYEES

Output5:

	LAST_NAME	SALARY	12*(SALARY+100)
1	King	24000	289200
2	Yang	17000	205200
3	Garcia	17000	205200
4	James	9000	109200
5	Miller	6000	73200
6	Williams	4800	58800
7	Jackson	4800	58800
8	Nguyen	4200	51600

Defining a null value:

Query6:

Select last_name,job_id,salary,commission_pct
FROM HR.EMPLOYEES;

Output6:

	LAST_NAME	JOB_ID	SALARY	COMMISSION_PCT
1	King	AD_PRES	24000	(null)
2	Yang	AD_VP	17000	(null)
3	Garcia	AD_VP	17000	(null)
4	James	IT_PROG	9000	(null)
5	Miller	IT_PROG	6000	(null)
6	Williams	IT_PROG	4800	(null)
7	Jackson	IT_PROG	4800	(null)
8	Nguyen	IT_PROG	4200	(null)

NULL values in Arithmetic Expressions:

Query7:

Select last_name,12*salary*commission_pct
FROM HR.EMPLOYEES;

Output7:

Download ▾ Execution time: 0.007 seconds		
	LAST_NAME	12*SALARY*COMM
1	King	(null)
2	Yang	(null)
3	Garcia	(null)
4	James	(null)
5	Miller	(null)
6	Williams	(null)
7	Jackson	(null)
8	Nguyen	(null)

Using Column Alias:

Query8:

Select last_name AS name,commission_pct comm
FROM HR.EMPLOYEES;

Output8:

Download ▾ Execution time: 0.007 seconds		
	NAME	COMM
1	King	(null)
2	Yang	(null)
3	Garcia	(null)
4	James	(null)
5	Miller	(null)
6	Williams	(null)
7	Jackson	(null)
8	Nguyen	(null)

Query9:

Select last_name "NAME", salary*12 "Annual Salary"
FROM HR.EMPLOYEES;

Output9:

Download ▾ Execution time: 0.006 seconds		
	NAME	ANNUAL SALARY
1	King	288000
2	Yang	204000
3	Garcia	204000
4	James	108000
5	Miller	72000
6	Williams	57600
7	Jackson	57600
8	Nguyen	50400

Concatenation Operator:

Query10:

Select last_name || job_id AS "Employees"
FROM HR.EMPLOYEES;

Output10:

Download ▾ Execution time: 0.007 seconds		
	EMPLOYEES	
1	AbelSA_REP	
2	AndeSA_REP	
3	AtkinsonST_CLERK	
4	BaidaPU_CLERK	
5	BandaSA_REP	
6	BatesSA_REP	
7	BellSH_CLERK	
8	BernsteinSA_REP	

Using Literal Character Strings:

Query11:

Select last_name || 'is a' || job_id AS "Employee Details"
FROM HR.EMPLOYEES;

Output11:

EMPLOYEES	
1	AbelSA_REP
2	AndeSA_REP
3	AtkinsonST_CLERK
4	BaidaPU_CLERK
5	BandaSA_REP
6	BatesSA_REP
7	BellSH_CLERK
8	BernsteinSA_REP

Duplicate Rows:

Query12:

Select DISTINCT department_id
FROM HR.EMPLOYEES;

Output12:

DEPARTMENT_ID	
1	90
2	60
3	100
4	30
5	50
6	80
7	(null)
8	10